

$$\Psi(x,t) \rightarrow \Psi(x,y,z,t) = \Psi(\vec{r},t)$$

$$\Psi(x) \rightarrow \Psi(\vec{r})$$

$$dx \rightarrow dx dy dz = dV = r^2 dr d(\cos\Theta) d\phi$$

$$\text{norma: } \int \Psi^*(\vec{r},t) \Psi(\vec{r},t) dV = 1$$

$$\text{pričakovana vr. } \langle A \rangle = \int \Psi^*(\vec{r},t) \hat{A} \Psi(\vec{r},t) dV$$

$$\text{GK: } \hat{p} = -i\hbar \frac{\partial}{\partial x} \rightarrow \hat{\vec{p}} = (\hat{p}_x, \hat{p}_y, \hat{p}_z) = -i\hbar \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = -i\hbar \nabla$$

$$\text{Kin. E: } \hat{T} = \frac{\hat{\vec{p}}^2}{2m} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) = -\frac{\hbar^2}{2m} \nabla^2$$

$$\text{Celotna E: } \hat{H} = \hat{T} + \underbrace{\hat{V}(\vec{r})}_{\text{omejeno se samo na } V(\vec{r}), \text{ ne pa npr. } V(\vec{r},t)}$$

$$\text{NSE: } \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \Psi(\vec{r},t) = i\hbar \frac{\partial \Psi}{\partial t}$$

$$\text{SSE: } \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \Psi(\vec{r}) = E \Psi(\vec{r})$$

$$\text{Ravni val: } \Psi(\vec{r},t) = A e^{i \underbrace{(k_x x + k_y y + k_z z)}_{\vec{k} \cdot \vec{r}} - \omega t}$$

• lastno stanje za GK oblike $p_\alpha = \hbar k_\alpha$; $\alpha = \{x, y, z\}$

$$\text{in } E = \hbar \omega = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2)$$

∞ pot. jama v 3d

$$V(\vec{r}) = \begin{cases} 0; & 0 \leq x, y, z \leq a \\ \infty; & \text{sicer} \end{cases}$$

SSE:

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi(x, y, z) = E \psi(x, y, z)$$

To reši nastavek:

$$\psi(x, y, z) = A \sin(k_x x) \sin(k_y y) \sin(k_z z)$$

$$\xrightarrow{\text{SSE}} E = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2)$$

k_x, k_y, k_z dobimo iz robnih pogojev:

$$\psi(0, y, z) = \psi(a, y, z) = \psi(x, 0, z) = \dots = 0$$

$$\rightarrow k_\alpha = \frac{n_\alpha \pi}{a}; \alpha \in \{x, y, z\}$$

$$\rightarrow E_{\underbrace{n_x n_y n_z}_{3 \text{ kv. št.}}} = \frac{\hbar^2 \pi^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2)$$



Stanja so DEGENERIRANA: razl. stanja z razl. kv.št. imajo enake E

$$\text{npr.: } E_{2,1,1} = E_{1,2,1} = E_{1,1,2}$$

to se je zgodilo, ker so vse stranice = a

• če porušimo simetrijo, degeneracija (delne) izgine

npr. namesto a, a, a

$$\text{imamo } a, b, c: \quad k_x = \frac{n_x \pi}{a}, \quad k_y = \frac{n_y \pi}{b}, \quad k_z = \frac{n_z \pi}{c}$$

$$\rightarrow E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right)$$

$$a=b=c$$

$$a < b < c$$

$$\epsilon_{221} = \epsilon_{212} = \epsilon_{122} = 9\epsilon_1$$

$$\epsilon_{211} = \epsilon_{121} = \epsilon_{112} = 6\epsilon_1$$

$$\epsilon_{111} = 3\epsilon_1$$

$$\epsilon_{111}$$

Harmonski oscilator

$$V(x) = \frac{1}{2} kx^2 \rightarrow \frac{1}{2} k(x^2 + y^2 + z^2)$$

$$SSE: -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi + \frac{1}{2} k(x^2 + y^2 + z^2) \psi = E \psi$$

Robni pogoji:

$$\psi(x, y, z) \rightarrow 0 \text{ ko } x, y, z \rightarrow 0$$

$$\text{Nastavek } \psi(x, y, z) = A u(x) v(y) w(z)$$

$$\rightarrow -\frac{\hbar^2}{2m} \left(\frac{\partial^2 u}{\partial x^2} v w + \frac{\partial^2 v}{\partial y^2} u w + \frac{\partial^2 w}{\partial z^2} u v \right) + \frac{1}{2} k(x^2 + y^2 + z^2) u v w = E u v w$$

$$\underline{-\frac{\hbar^2}{2m} \frac{\partial^2 u}{\partial x^2} \frac{1}{u}} - \underline{\frac{\hbar^2}{2m} \frac{\partial^2 v}{\partial y^2} \frac{1}{v}} - \underline{\frac{\hbar^2}{2m} \frac{\partial^2 w}{\partial z^2} \frac{1}{w}} + \underline{\frac{1}{2} k(x^2 + y^2 + z^2)} = E$$

♦ - odv. samo od x

♦ - ... y

$$\lambda_s = -\frac{\hbar^2}{2m} \frac{1}{t} \frac{\partial^2 t}{\partial s^2} + \frac{1}{2} k s^2; \quad s = \{x, y, z\}$$

$$t = \{u, v, w\}$$

$$\lambda_x + \lambda_y + \lambda_z = E$$

$$1d: E = \hbar \omega \left(n_\alpha + \frac{1}{2} \right) \quad \alpha \in \{x, y, z\}$$

$$3d: E = \hbar \omega \left(n_x + n_y + n_z + \frac{3}{2} \right); \quad n_x, n_y, n_z = 0, 1, \dots$$

Tudi tu imamo degeneracijo: $n_x, n_y, n_z = \begin{matrix} 1,0,0 \\ 0,1,0 \\ 0,1,1 \end{matrix} \left. \vphantom{\begin{matrix} 1,0,0 \\ 0,1,0 \\ 0,1,1 \end{matrix}} \right\} \text{ vse imajo enako } \epsilon$

Nedoločnost in komutatorji operatorjev

komutator $\hat{A}, \hat{B}$: $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$

če $[\hat{A}, \hat{B}] = 0$, $\hat{A}$ in $\hat{B}$ komutirata

• njihov vrstni red ni pomemben

zgleď (1d):

$$\hat{x} = x$$

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}$$

$$[\hat{x}, \hat{p}]\psi = (\hat{x}\hat{p} - \hat{p}\hat{x})\psi = -i\hbar \left(x \frac{\partial \psi}{\partial x} - \frac{\partial}{\partial x} (x \psi) \right) =$$

$$= -i\hbar \left(x \frac{\partial \psi}{\partial x} - \frac{\partial x}{\partial x} \psi - x \frac{\partial \psi}{\partial x} \right) = \underline{\underline{i\hbar \psi}}$$

ker ψ poljuben:

$$[\hat{x}, \hat{p}] = i\hbar \rightarrow \hat{x}, \hat{p} \text{ ne komutirata}$$

Če operatorja NE komutirata, ne $\exists$ stanje, v katerem bi bili količini ostro določeni.

• če bi takšno stanje $\exists$:

ψ_{AB} ↙

$$\left. \begin{aligned} \hat{A}\psi_{AB} = A\psi_{AB} &\rightarrow \hat{B}\hat{A}\psi_{AB} = \hat{B}A\psi_{AB} = AB\psi_{AB} \\ \hat{B}\psi_{AB} = B\psi_{AB} &\rightarrow \hat{A}\hat{B}\psi_{AB} = \hat{A}B\psi_{AB} = BA\psi_{AB} \end{aligned} \right\} \rightarrow [\hat{A}, \hat{B}]\psi_{AB} = (AB - BA)\psi_{AB} = 0$$

$\Rightarrow$ protislovje

Nedoločnosti: $\delta\hat{A} = \hat{A} - \langle A \rangle$

$\delta\hat{B} = \hat{B} - \langle B \rangle$

ni težko pokazati:

$$\boxed{\delta A \delta B \geq \frac{1}{2} |\langle [\delta\hat{A}, \delta\hat{B}] \rangle| = \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|}$$

Primer: $\hat{A} = \hat{x}$
 $\hat{B} = \hat{p}_x \rightarrow \underline{\delta x \delta p_x \geq \frac{\hbar}{2}}$

Toda: $[\hat{x}, \hat{p}_y] = 0 \rightarrow \delta x \delta p_y \geq 0$

Kvantizacija vrtilne količine

klasično: $\vec{L} = \vec{r} \times \vec{p}$ TIRNA VK

kvantno: $\hat{L} = (\hat{L}_x, \hat{L}_y, \hat{L}_z) = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y}, z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z}, x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$

$$[\hat{L}_x, \hat{L}_y] = -\hbar^2 \left[\left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) - \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \right] =$$

$$= -\hbar^2 \left[y \frac{\partial}{\partial x} + z \frac{\partial^2}{\partial x \partial z} - z^2 \frac{\partial^2}{\partial x \partial y} - xy \frac{\partial^2}{\partial z^2} + xz \frac{\partial^2}{\partial y \partial z} \right.$$

$$\left. - yz \frac{\partial^2}{\partial x \partial z} + z^2 \frac{\partial^2}{\partial x \partial y} + xy \frac{\partial^2}{\partial z^2} - xz \frac{\partial^2}{\partial y \partial z} - x \frac{\partial}{\partial y} \right] =$$

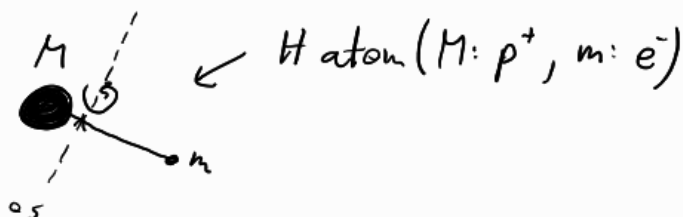
$$= i\hbar \underbrace{\left(-i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \right)}_{\hat{L}_z} = i\hbar \hat{L}_z$$

$$\begin{aligned} [\hat{L}_x, \hat{L}_y] &= i\hbar \hat{L}_z \\ [\hat{L}_y, \hat{L}_z] &= i\hbar \hat{L}_x \\ [\hat{L}_z, \hat{L}_x] &= i\hbar \hat{L}_y \end{aligned}$$

$\Rightarrow$ stanje, ki je lastno za vse 3 kom. VK hkrati

To je relevantno za opis ROTATORJA

npr. ročka



$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

$$\rightarrow [\hat{L}^2, \hat{L}_x] = [\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \hat{L}_x] = \cancel{\hat{L}_x^2} + \hat{L}_y^2 \hat{L}_x + \hat{L}_z^2 \hat{L}_x - \cancel{\hat{L}_x^2} - \hat{L}_x \hat{L}_y^2 - \hat{L}_x \hat{L}_z^2$$

$$\hat{L}_z^2 \hat{L}_x = \hat{L}_z \hat{L}_z \hat{L}_x = \hat{L}_z (\hat{L}_x \hat{L}_z - i\hbar \hat{L}_y) = \hat{L}_z \hat{L}_x \hat{L}_z - i\hbar \hat{L}_z \hat{L}_y = \hat{L}_x \hat{L}_z^2 + i\hbar (\hat{L}_y \hat{L}_z + \hat{L}_z \hat{L}_y)$$

$$[\hat{L}_x, \hat{L}_z] = \hat{L}_x \hat{L}_z - \hat{L}_z \hat{L}_x = i\hbar \hat{L}_y$$

analogno $\hat{L}_x \hat{L}_y^2 = \hat{L}_y^2 \hat{L}_x - i\hbar (\hat{L}_y \hat{L}_z + \hat{L}_z \hat{L}_y)$

$$\rightarrow [\hat{L}^2, \hat{L}_x] = \cancel{\hat{L}_y^2 \hat{L}_x} + \cancel{\hat{L}_x \hat{L}_z^2} + i\hbar (\cancel{\hat{L}_y \hat{L}_z} + \cancel{\hat{L}_z \hat{L}_y}) - \cancel{\hat{L}_y^2 \hat{L}_x} - \cancel{\hat{L}_x \hat{L}_z^2} - i\hbar (\cancel{\hat{L}_y \hat{L}_z} + \cancel{\hat{L}_z \hat{L}_y}) = 0$$

$\Rightarrow$ hkrati lahko ostro dobimo kvadrat VK ($\hat{L}^2$) in eno komp. VK (L_α ; $\alpha = \{x, y, z\}$)

• običajno izberemo $\hat{L}^2, L_z$, os z je poimenovana kvantizacijska os (in je najrelevantnejša)

Rotator: raje v sferične koord.

$$\Theta, \varphi, r$$

- iščemo VF za opis rotatorja oz. verj. gostoto za napoved orientacije ročke oz. njegove tirne VK v bližini Θ, φ

$$x = r \sin \Theta \cos \varphi$$

$$y = r \sin \Theta \sin \varphi$$

$$z = r \cos \Theta$$

$$\hat{L}_x = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) = -i\hbar \left(-\sin \varphi \frac{\partial}{\partial \Theta} - \cot \Theta \cos \varphi \frac{\partial}{\partial \varphi} \right)$$

$$\hat{L}_y = -i\hbar \left(\cos \varphi \frac{\partial}{\partial \Theta} - \cot \Theta \sin \varphi \frac{\partial}{\partial \varphi} \right)$$

$$\boxed{\hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi}} \text{ zelo preprosto}$$

$$\boxed{\hat{L}^2 = -\hbar^2 \left(\frac{\partial^2}{\partial \Theta^2} + \cot \Theta \frac{\partial}{\partial \Theta} + \frac{1}{\sin^2 \Theta} \frac{\partial^2}{\partial \varphi^2} \right)}$$

$$[\hat{\varphi}, \hat{L}_z] = i\hbar$$

$$\downarrow$$

$$-i\hbar \frac{\partial}{\partial \varphi}$$

$$\uparrow$$

analogno $[\hat{x}, \hat{p}_x] = i\hbar$

$$\downarrow$$

$$-i\hbar \frac{\partial}{\partial x}$$

Heisenberg:

$$\Delta \varphi \Delta L_z \geq \frac{\hbar}{2}$$

Lastne funkc. operatorja $\hat{L}_z$, oznaka $\Phi(\varphi)$

$$\hat{L}_z \Phi = -i\hbar \frac{d\Phi}{d\varphi} \stackrel{?}{=} L_z \Phi$$

$$\Rightarrow \Phi = A e^{im\varphi}$$

$$L_z = m\hbar, m = 0, \pm 1, \pm 2, \dots$$

$$\Phi(\varphi) = \Phi(\varphi + 2\pi) \Rightarrow m \in \mathbb{Z}$$

normalizacija: $1 = \int_0^{2\pi} |\Phi|^2 d\varphi = \int_0^{2\pi} |A|^2 e^{im\varphi} e^{-im\varphi} d\varphi \Rightarrow A = \frac{1}{\sqrt{2\pi}}$

$$\Rightarrow \boxed{\Phi_m(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi}}$$

kvantno št. tretje (z) komponente tirne VK
(oz. projekcije na os z)

Lastne funkcije $\hat{L}^2$ (oznaka $\gamma(\theta, \varphi)$)

$$\hat{L}^2 \gamma = -\hbar^2 \left(\frac{\partial^2 \gamma}{\partial \theta^2} + \cos \theta \frac{d\gamma}{d\theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2 \gamma}{\partial \varphi^2} \right) = \lambda \gamma$$

$$\left/ \frac{\sin^2 \theta}{\gamma} \right.$$

na skvele: $\gamma(\theta, \varphi) = \Theta(\theta) \cdot \Phi(\varphi)$

$$\hbar^2 \frac{1}{\Theta} \sin^2 \theta \frac{d^2 \Theta}{d\theta^2} + \hbar^2 \frac{1}{\Theta} \cos \theta \sin \theta \frac{d\Theta}{d\theta} + \lambda \sin^2 \theta = - \frac{1}{\Phi} \frac{d^2 \Phi}{d\varphi^2} \hbar^2 \stackrel{\text{od prej}}{=} m^2 \hbar^2$$

za $\Theta(\theta)$ je bolj sitno:

$$\lambda = \hbar^2 L(L+1); L = 0, 1, 2, \dots$$

$$|m| \leq L$$

→ lastne funkc. γ :

$$\boxed{\gamma_{lm}(\theta, \varphi) = A_{lm} P_l^m(\cos \theta) e^{im\varphi}}$$

krogelne funkcije (sferični harmoniki)

→ pridruženi Legendra polinomi:

$$P_l^m(t) = (1-t^2)^{m/2} \frac{d^m}{dt^m} (P_l(t))$$

navadni Leg. polinomi

$$\hat{L}^2 Y_{lm} = \underbrace{\hbar^2 l(l+1)}_{\hat{L}^2} Y_{lm}$$

$$\hat{L}_z Y_{lm} = \underbrace{\hbar m}_{L_z} Y_{lm}$$

↖ kš projekcije tirne VK na z-os
 $|m| \leq l$
 $l = 0, 1, \dots$
 ↖ kš velikosti tirne VK

nekaj prvih:

$$Y_{00} = \frac{1}{\sqrt{4\pi}} \text{ (točka kaže kamorkoli v prostoru)}$$

$$Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta$$

$$Y_{1\pm 1} = \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\varphi}$$

$$Y_{20} = \sqrt{\frac{5}{16\pi}} (3\cos^2 \theta - 1)$$

$$Y_{2\pm 1} = \sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{\pm i\varphi}$$

$$Y_{2\pm 2} = \sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i\varphi}$$

⋮

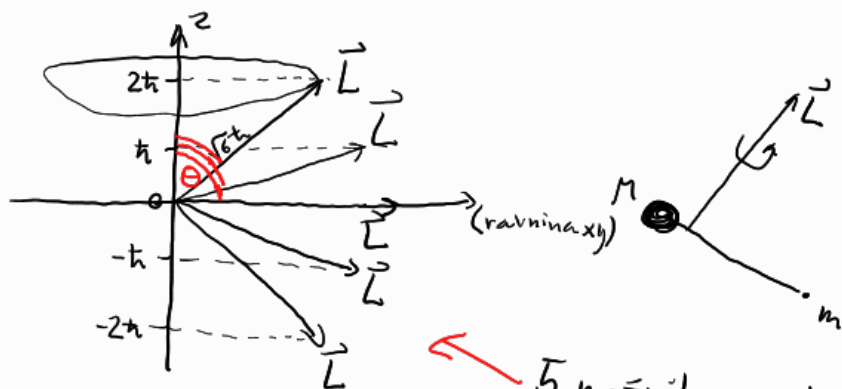
Lastnosti:

• norma $\int_0^{2\pi} \int_0^\pi Y_{lm}^* Y_{lm} \sin \theta d\theta d\varphi = 1$

• ortogonalnost $\iint Y_{l'm'}^* Y_{lm} \sin \theta d\theta d\varphi = \delta_{l'l} \delta_{m'm}$

• vsako $f(\theta, \varphi)$ lahko razvijemo po Y_{lm} : $f = \sum_{lm} c_{lm} Y_{lm}$

Zgled: $l=2 \rightarrow |\vec{L}|^2 = \hbar^2 l(l+1) = 6\hbar^2 \rightarrow |\vec{L}| = \sqrt{6} \hbar$



↖ 5 možnih projekcij L_z
 vektorja $\vec{L}$ ($l=2; m=\{0, \pm 1, \pm 2\}$)

Degeneracija je $2l+1$

VK je kvantizirana:

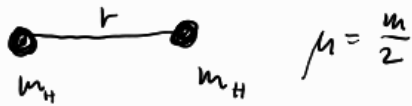
$$\cos \theta = \frac{L_z}{|\vec{L}|} = \frac{m\hbar}{\hbar \sqrt{l(l+1)}} \rightarrow l=2, m=2: \theta \approx 35,5^\circ$$

$$E_{\text{rot}} = \frac{\vec{L}^2}{2J} = \frac{\hbar^2 l(l+1)}{2\mu r^2}$$

$$\mu = \frac{mM}{m+M} \text{ reducirana masa}$$

r ... razdalja med m in M

Primer: H_2



$$\frac{\hbar^2}{2J} = \frac{\hbar^2}{m_H r^2} \approx \underbrace{2 \cdot 10^{-3}}_{\text{tipične rot. energije}} \text{ eV}$$